

SEUNGJOO LEE

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KEYWORDS

Machine Learning for Healthcare, Mobile Sensing, Wearable, Foundation Models, Privacy-Preserving ML

EXPERIENCE

Carnegie Mellon University, *Ph.D. Student*. Advisor: [Mayank Goel](#) & [Justin Chan](#) Aug 2025 – Present

- Aligning **LLMs** with **wearable sensor data** for personalized healthcare applications.
- Developing **privacy-preserving interaction systems** with the physical world for **AR/MR**.

Microsoft Research, *Research Intern*. Mentor: [Lili Qiu](#) Sep 2024 – Mar 2025

- Developed an **earable-form wearable device** for in-the-wild ExG sensing and a **task-agnostic representation learning framework** trained on free-living ExG data, enabling robust performance across diverse downstream tasks (gaze tracking, audio & visual interest, smell & taste) – [ICLR'26 \[C4\]](#)

KAIST, *Graduate Student*. Advisor: [Sung-Ju Lee](#) Dec 2018 – Jul 2025

- Enabled **99.9% label-free federated learning** by introducing a semi-supervised FL approach that performs reliably with extremely limited labeled data. – [NeurIPS'24 \[C3\]](#)
- Developed an **on-device mental-health monitoring system** that analyzes speech and text data on smartphones with **NLP & AI**, achieving a 0.15 AUROC gain and 8.21% MAE reduction over prior methods. – [EMNLP'23 Main \[C2\]](#)
- Implemented **on-device training** for LLMs (BERT, LLaMA-7B) on smartphones using PEFT, LoRA, and quantization, reducing trainable parameters by ~99% for efficient fine-tuning.
- Invented an **eyeglass attachable device for eating detection** with **multimodal sensors** (IMU and piezo) with AI, achieving 98% accuracy and increasing the battery life by 4.03× over state-of-the-art. – [CHI'22 Best Paper \[C1\]](#)
- Developed a UWB-based system to **detect approaching personal-mobility vehicles** and warn pedestrians to prevent collisions — [MobiSys'25 Workshop \[W1\]](#)

Samsung Electronics, *Intern* Jul 2021 – Aug 2021
MX (Mobile Experience) Division, AI Server Development Team

- Developed a RESTful server using Spring Boot; designed server APIs and implemented CI/CD pipelines in a production engineering environment. Selected as the **Best Project** among 200+ interns.
- Proposed a business concept for elderly-friendly kiosk services and delivered division-wide presentation; Won **1st place** in Creathon competition.

SK Hynix, *Intern* Dec 2017 – Jan 2018
NAND Development Branch, Solution Algorithm Division

- Optimized NAND simulation code, achieving **66% performance improvement** over the existing implementation; awarded **Best Project** in the division.

EDUCATION

Carnegie Mellon University Aug 2025 – Present
School of Computer Science, Ph.D. in Societal Computing Advisor: [Mayank Goel](#) & [Justin Chan](#)

KAIST (Korea Advanced Institute of Science and Technology) Sep 2022 – Jul 2025
M.S. and Ph.D. in Electrical Engineering (Transferred to Carnegie Mellon University) Advisor: [Sung-Ju Lee](#)

KAIST (Korea Advanced Institute of Science and Technology) Mar 2016 – Aug 2022
B.S. in Computer Science * Including 1.5 years of mandatory military service

🏆 Graduated with Honors (2nd place out of 715 students)

🏆 Summa Cum Laude (Major GPA : 4.13/4.3 | Total GPA : 4.0/4.3)

PUBLICATIONS

(C: Conference, D: Demo, W: Workshop, B: Book, Pr: Preprint)

* *Equal contribution.* † *Equal senior role*

- [C4] **Beyond Hearing: Learning Task-agnostic ExG Representations from Earphones via Physiology-informed Tokenization**
Hyungjun Yoon*, **Seungjoo Lee***, Yu Wu*, Xiaomeng Chen*, Taiting Lu, Freddy Yifei Liu, Taekyung Lee, Hyeongheon Cha, Haochen Zhao, Gaoteng Zhao, Sung-Ju Lee, Dongyao Chen, Cecilia Mascolo, Lili Qiu
International Conference on Learning Representations (ICLR) 2026
Top ML Conference - Acceptance rate 28.2%
- [C3] **(FL)²: Overcoming Few Labels in Federated Semi-Supervised Learning**
Seungjoo Lee, Thanh-Long V. Le, Jaemin Shin, Sung-Ju Lee
Conference on Neural Information Processing Systems (NeurIPS) 2024
Top ML Conference - Acceptance rate 25.8%
- [C2] **FedTherapist: Mental Health Monitoring with User-Generated Linguistic Expressions on Smartphones via Federated Learning**
Jaemin Shin, Hyungjun Yoon, **Seungjoo Lee**, Sungjoon Park, Yunxin Liu, Jinho D. Choi, Sung-Ju Lee
Conference on Empirical Methods in Natural Language Processing (EMNLP Main) 2023
Top NLP Conference - Acceptance rate 21.3%
- [C1] **MyDJ: Sensing Food Intakes with an Attachable on Your Eyeglass Frame**
Jaemin Shin, **Seungjoo Lee**, Taesik Gong, Hyungjun Yoon, Hyunchul Roh, Andrea Bianchi, Sung-Ju Lee
Conference on Human Factors in Computing Systems (CHI) 2022
🏆 **Best Paper Honorable mention award (top 5%)**
Top HCI Conference - Acceptance rate 24.7%
- [W1] **CrashSniffer: UWB-Based Anchor-Free Pedestrian Collision Prediction for Personal Mobility Vehicles**
Taekyung Lee, Juseung Lee, Ryuhaerang Choi, **Seungjoo Lee**, Hyeongheon Cha, Hyungjun Yoon, Song Min Kim, Sangwook Bak, Sung-Ju Lee
MobiSys Workshops (EnvSys) 2025
- [D1] **Accurate Eating Detection on a Daily Wearable Necklace (Demo)**
Jaemin Shin, **Seungjoo Lee**, Sung-Ju Lee
International Conference on Mobile Systems, Applications and Services (MobiSys) 2019
- [B1] **High School Arduino**
Comprehensive guide on conducting scientific experiments using Arduino, **commercially published and sold in major bookstores (4,100+ copies sold)**.
Chapters 4, 7, and 13 authored by me.

AWARDS & HONORS

- NeurIPS 2025 Top Reviewer (Top 8.02%)** Oct 2025
Recognized among the top 8.02% of 24,429 reviewers
- CHI 2022 Honorable Mention Award** Mar 2022
Awarded to top 5% of all submissions in CHI 2022 [C1].
- Winner of the Graduate of the Year Award (KAIST Board of Trustee Chairpeson's Prize)** Feb 2023
2nd place out of 715 students, awarded to top 5 students (0.7%) who demonstrated outstanding performances in various activities as well as in grades. Awarded at the commencement ceremony.
- Summa Cum Laude** Sep 2022
Awarded for achieving the highest academic performance among students.
- Dean's List** Spring 2019, Fall 2019
Awarded to top 3% among 2,900+ students in College of Engineering.
- Engineering Innovator Award** Sep 2022
Awarded to students showing outstanding performance in extracurricular activities, including academic publications,

entrepreneurial activities, exhibitions, and inventions
Five students are picked from college of engineering (2,900+) each semester.

KAIST Breakthroughs

Spring 2023

Featured in biannual KAIST webzine showcasing groundbreaking works [C2].

U.S. Army Certificate of Appreciation

Jul 2021

Awarded for exemplary service during military duty, serving as a role model for others.

National Excellence Scholarship

Fall 2018 – Fall 2019

National scholarship to students who showed excellence.

Samsung Humantech Paper Award

Spring 2015

Fourth prize on "An optimal path of navigation based on fractal dimension".

PATENTS

- [P5] "Wearable Devices for Status Sensing", Lili Qiu, Hyungjoon Yoon, Seungjoo Lee, Xiaomeng Chen, Yu Wu, Taiting Lu, Yifei Liu, Zilong Wang, *US patent (In progress)*
- [P4] "UWB-based Pedestrian Collision Avoidance System for Personal Mobility Devices", Sung-Ju Lee, Taeckyoung Lee, Hyungjun Yoon, Seungjoo Lee, Hyeongheon Cha, *Korea patent (Filing date: 2025.02.12, No. 10-2025-0018095)*
- [P3] "Adaptive State-Space Model-Based Deep Learning System for Real-Time Data Inference on Mobile Devices", Sung-Ju Lee, Seungjoo Lee, HyungJun Yoon, *Korea patent (Filing date: 2025.03.31, No. 10-2025-0040898)*
- [P2] "System and Method for Monitoring Mental Health Based on Smartphone User Language Expressions Through Federated Learning", Sung-Ju Lee, HyungJun Yoon, Seungjoo Lee, Jaemin Shin, *Korea patent (Filing date: 2024.10.04, No. 10-2024-0135031)*
- [P1] "Artificial Intelligence Model Training Method and Apparatus for Voice Phishing Detection", Sung-Ju Lee, Seungjoo Lee, HyungJun Yoon, *Korea patent (Filing date: 2023.10.04, No. 10-2023-0131748)*

INVITED TALKS

MSRA Intern Tech Talk, MSRA

Nov 2024

Overcoming Few Labels in Federated Semi-Supervised Learning [C3]

ACADEMIC SERVICES

Conference Reviewer

- International Conference on Learning Representations (**ICLR**), 2026
- Conference on Neural Information Processing Systems (**NeurIPS**), 2025 – Top Reviewer
- International Conference on Mobile Systems, Applications, and Services (**MobiSys**), 2023–2025
- International Conference On Mobile Computing And Networking (**MobiCom**), 2023–2025
- Special Interest Group on Data Communication (**SIGCOMM**), 2023
- International Joint Conference on Pervasive and Ubiquitous Computing (**UbiComp**), 2022

Outreach

- **Educational Support for Students in Difficult Circumstances** Mar 2020 – Jun 2021
Taught math and science to 4 middle school students for 2 hours every week (total ~136 hours)
- **Mentor for the Gifted Education Program, SW/AI Camp** Summer 2022
Facilitated a 2-night, 3-day program for elementary and middle school students, focusing on creating SW/AI-related kits, providing career mentoring, and conducting quizzes
- **CS101 Tutor** Spring 2017, Fall 2017, Spring 2022
Taught 5 freshman studying CS101 (introduction to programming)

TEACHING EXPERIENCES

Computer Network (EE323) Head of teaching assistants (Prof. Sung-Ju Lee)	Spring 2024
Operating Systems and System Programming for Electrical Engineering (EE415) Teaching assistant (Prof. Sung-Ju Lee)	Fall 2023
Programming Structures for Electrical Engineering (EE209) Teaching assistant (Prof. Sung-Ju Lee)	Spring 2023

MENTORING EXPERIENCES

Seoyoung Park , Undergraduate research intern at NMSL Doing a research project about <i>federated learning</i> together	Jun 2024 – Oct 2024
Thanh Long Le Viet , Undergraduate research intern at NMSL Doing a research project about <i>federated learning</i> together	Oct 2023 – May 2024
Rachel Kim , Undergraduate research intern at NMSL Doing a research project about <i>UWB-based personal mobility warning system for pedestrians</i> together	Dec 2023 – Feb 2024

MILITARY SERVICE

KATUSA (Korean Augmentation to the U.S. Army) Served national service at eighth army as a KATUSA, providing translation between the U.S. and Korean army. Selected to participate as a Korean military interpreter in the EIB (Expert Infantryman Badge) contest, the U.S. Army's prestigious competition.	Dec 2019 - Jun 2021
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EXTRA-CURRICULAR ACTIVITIES

International Students Organization (ISO) Introduce Korean culture to international students.	Sep 2016 - Dec 2019
Flute Player in Cantabile (Orchestra)	Mar 2014 - Jan 2016
Embedded Software Club Been a club leader for a year. Published book about doing scientific experiments using Arduino [B1].	Mar 2014 - Dec 2015